THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 4: **Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A magnet is brought near an unknown rod. It is attracted at both ends. The rod is: 1

(a) either a magnet or magnetic material **(b)** surely non-magnetic
(c) plastic **(d)** surely a magnet

2. Which of the following correctly lists only magnetic materials? 1

(a) iron, nickel, cobalt **(b)** copper, silver, gold
(c) wood, rubber, glass **(d)** water, oil, air

3. In a toy, two magnets are placed with their like poles facing; the toy parts will: 1

(a) stick together **(b)** melt
(c) spin **(d)** push away from each other

4. A rod repels the north pole of a magnet. The end of the rod near the north pole is: 1

(a) non-magnetic **(b)** neutral **(c)** south pole **(d)** north pole

5. A student says 'All metals are attracted by a magnet'. Which metal shows this is wrong? 1

(a) cobalt **(b)** nickel **(c)** iron **(d)** aluminium

6. A toy train car joins another with magnets. The ends that join must be: 1

(a) like poles **(b)** neutral **(c)** made of wood **(d)** unlike poles

7. Two magnets attract when their poles face each other. The poles are: 1

(a) like **(b)** unlike **(c)** both north **(d)** both south

8. A steel spoon is attracted but a copper coin is not. This shows: 1

(a) no metal is magnetic **(b)** all metals are magnetic
(c) copper is stronger **(d)** only some metals are magnetic

9. Which of these is the safest place to keep a bar magnet? 1

(a) near a mobile phone **(b)** in the sun
(c) on a stove **(d)** in a box with a keeper

10. A bar magnet suspended by a thread rests with its marked end pointing north. That end is its: 1

(a) centre **(b)** north-seeking pole
(c) neutral **(d)** south pole

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A student finds that a rod attracts both poles of a magnet. Can the rod be a magnet? Explain. 2

12. What would happen if two south poles are brought together? 2

13. Which is stronger, a magnet's pull at the pole or at the middle? Explain. 2

14. How can you tell whether a bar of iron or a magnet has been given to you without other tools but a magnet? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student has a mix of iron filings, sand and salt. How to separate all three? 3

16. How can the poles of an unmarked bar magnet be identified with another marked magnet? 3

17. Name three non-magnetic metals and a use. 3

OR

Explain how a compass can be made with a needle, a cork and water. 3

18. Why does a suspended bar magnet oscillate before settling? 3

OR

What happens to a magnetic compass inside a car? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe an activity to show the repulsion between like poles and attraction between unlike poles and write the conclusion. 5

OR

Compare the behaviour of a magnet and an iron piece when brought near a bar magnet. 5

20. Explain with reasons: 5
(a) a magnet attracts iron pins through paper;
(b) a magnet does not attract copper coins;
(c) magnets are stored in pairs.

OR

Explain how iron filings can be used to find the shape of the magnetic field and its strength. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND